

Building Equivalent Ratio Tables

Answer Key

Grade 6–7 Mathematics

Quick Answers

1. 10, 15, 25	4. 1.5	7. 12, 25	10. 70
2. 7, 56	5. —	8. 37.5	11. 7, 63
3. 10, 15	6. 1.5, 30	9. —	12. 6, 24, 42

Common wrong answers

1. *If they got 13: added the difference between the two flour amounts to the milk instead of multiplying by the scale factor*
6. *If they got 24: added the difference 4 to the second entry instead of multiplying by the scale factor*
12. *If they got 16.5: split the 66 ounces by dividing by each ratio number separately instead of by the 11 total parts*
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1. Muffins: 2 cups flour to 5 cups milk; complete the milk row for 4, 6 and 10 cups of flour.

Every column must be the starting column times one number. Read the scale factor off the flour row, then apply the same factor to the milk entry.

$4 = 2 \times 2$, so milk $= 5 \times 2 = \boxed{10}$. $6 = 2 \times 3$, so milk $= 5 \times 3 = \boxed{15}$. $10 = 2 \times 5$, so milk $= 5 \times 5 = \boxed{25}$.

Grading note: 13 in the last cell is the additive error — the student saw flour go up by 8 and added 8 to the milk. Point at the second column: 4 is not $2 + 8$, it is 2×2 , and the same operation has to be used on both rows.

2. Beads: 3 red to 8 white; a row has 21 red.

Find the multiplier first, from the row where both entries are known:

$$21 \div 3 = 7$$

so the scale factor is $\boxed{7}$. Apply it to the partner entry: $8 \times 7 = 56$.

$\boxed{56}$ white beads

3. 5 teachers per 90 students; complete for 180 and 270 students.

This time the known row is the *student* row, so scale from there. $180 \div 90 = 2$, so teachers $= 5 \times 2 = \boxed{10}$.

$270 \div 90 = 3$, so teachers $= 5 \times 3 = \boxed{15}$.

The direction does not matter — what matters is that the same factor is used on both rows of a column.

4. 6 bottles every 4 seconds; find the unit rate.

The unit-rate column is the one whose seconds entry is 1. Divide both entries of the known column by 4:

$$\frac{6}{4} = \frac{3}{2}$$

1.5 bottles per second

5. Is 4 : 6 equivalent to 14 : 21?

Write each as a fraction and reduce:

$$\frac{4}{6} = \frac{2}{3}, \quad \frac{14}{21} = \frac{2}{3}$$

Both tables are built from the same ratio 2 : 3, so the tables are equivalent.

$4/6 = 14/21$

Grading note: the cross-product check is the same statement: $4 \times 21 = 84 = 6 \times 14$.

6. Video: 8 minutes to 20 megabytes; next column is 12 minutes.

The scale factor need not be a whole number. Divide the new entry by the old: $k = 12 \div 8 =$

1.5

Apply it to the storage row: $20 \times 1.5 = 30$.

30 megabytes

Grading note: 24 means the student added the difference $12 - 8 = 4$ to 20. That keeps the *difference* constant, not the ratio, and it is the single most common ratio-table error.

7. Paint: 3 blue to 5 yellow; find the two missing entries.

Left blank (yellow is 20): $20 \div 5 = 4$, so blue = $3 \times 4 = 12$.

Right blank (blue is 15): $15 \div 3 = 5$, so yellow = $5 \times 5 = 25$.

Blue: 12 Yellow: 25

Each column is scaled from the *first* column, not from its neighbour — though scaling from the neighbour works too if the factor is computed correctly.

8. Train: 150 miles in 4 hours; find the speed.

The unit-rate column has hours = 1, so divide both entries by 4:

$$\frac{150}{4} = 37.5$$

Since the hours entry is now 1, the miles entry *is* the speed in miles per hour.

37.5 mph

9. Lemonade: recipe P is 3 scoops to 5 cups, recipe Q is 5 scoops to 8 cups. Which is stronger?

“Stronger” means more mix per cup, so compare the scoops-per-cup unit rates — or, equivalently, the two fractions with a common denominator:

$$\frac{3}{5} = \frac{24}{40}, \quad \frac{5}{8} = \frac{25}{40}$$

$\boxed{3/5 < 5/8}$, so **recipe Q** is the stronger mix.

Grading note: “Q has more scoops, so Q is stronger” is right here but for the wrong reason — Q also has more water. Insist on the unit rate.

10. A table built from $5 : 3$ has a column reading x to 42.

Equivalent columns mean equal ratios, so set the two fractions equal:

$$\frac{x}{42} = \frac{5}{3}$$

Cross-multiply (or notice $42 = 3 \times 14$, so the factor is 14):

$$3x = 5 \times 42 = 210 \quad \implies \quad x = \frac{210}{3}$$

$$\boxed{x = 70}$$

Check with the scale factor: $5 \times 14 = 70$ and $3 \times 14 = 42$.

11. A table starts at 9 to 12; a later column has 84 in the second row. Find k and the missing entry.

The scale factor k multiplies *both* rows, so it satisfies

$$12k = 84 \quad \implies \quad k = \frac{84}{12}$$

$$\boxed{k = 7}$$

Now apply it to the first row: $9 \times 7 = 63$.

$$\boxed{63}$$

12. Synthesis: blue to yellow is $4 : 7$, and a batch totals 66 ounces.

Add a total row to the table. One “part-set” of the ratio is $4 + 7 = 11$ ounces, so the total row of the starting column is 11.

The batch’s total row reads 66, so the whole table was scaled by x where

$$11x = 66 \quad \implies \quad x = \frac{66}{11}$$

$$\boxed{x = 6}$$

Scale each row by 6: blue = 4×6 and yellow = 7×6 .

Blue: $\boxed{24}$ ounces Yellow: $\boxed{42}$ ounces

Check: $24 + 42 = 66$ ✓ and $24 : 42$ reduces to $4 : 7$ ✓.

Grading note: 16.5 ounces of blue means the student divided 66 by 4 — treating the ratio numbers as divisors instead of counting the 11 equal parts the total splits into.